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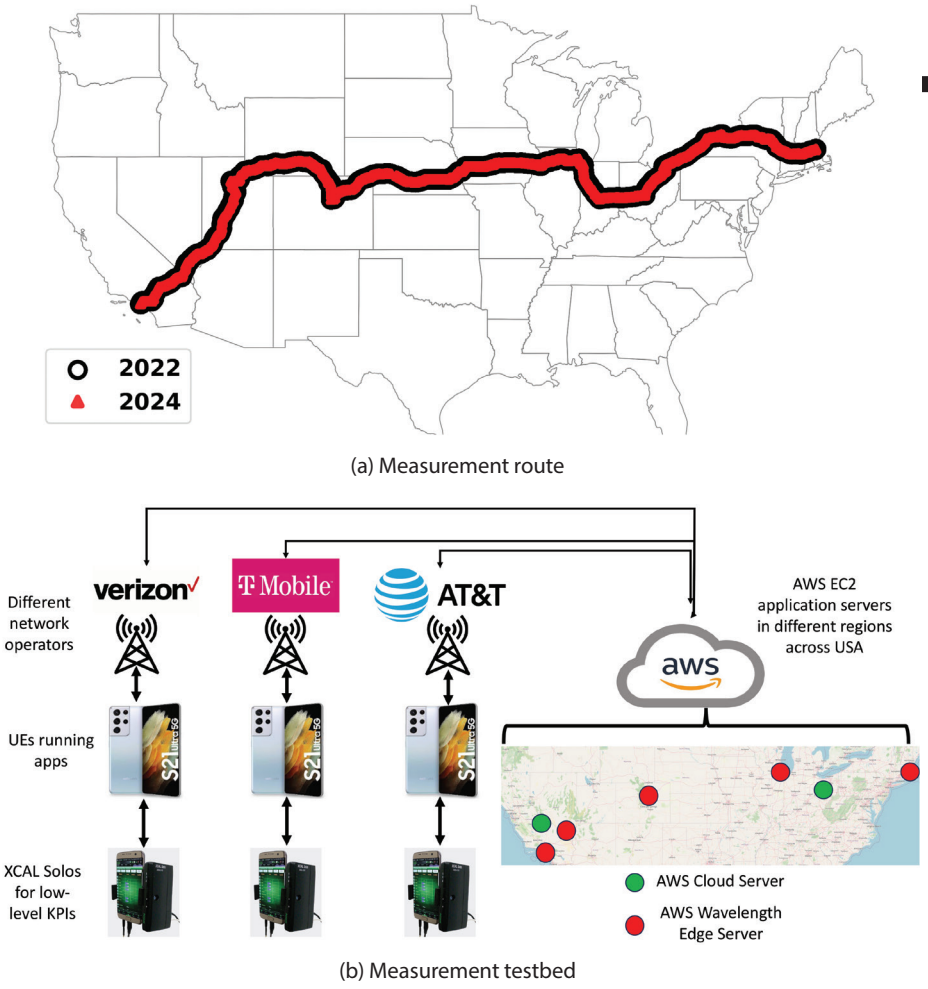


Roads, Radios, and Reality: What 5G Looks Like From the Driver's Seat

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When 5G first arrived, it came with big promises: ultra-high bandwidth, ultra-low latency, and a new generation of mobile experiences – from AR/VR to connected cars to early visions of the metaverse. Many of these applications were designed for the road, where consistency matters just as much as speed in maintaining high QoE. Yet the very design choices that give 5G its power – especially its use of higher-frequency spectrum – also make it fragile. Signals attenuate faster in higher frequency bands and coverage shrinks. And once you start moving, performance can vary wildly from one minute to the next.

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(a) Measurement route

(b) Measurement testbed

FIGURE 1. Measurement route and overview of our measurement testbed.

5G rollout started in 2019 and the wide-scale deployment has been rapid and aggressively marketed by all major mobile network operators. By now, with towers upgraded and new bands available across the country, one would expect cellular user experience to be dramatically better than during the LTE era, especially for the new class of latency-sensitive applications that LTE simply could not handle.

Researchers have tried to measure this progress. Many studies have offered snapshots of 5G performance, but often from static locations [5, 6, 7, 8, 9, 10, 11] or by focusing on narrow technical aspects such as handovers [13], or multi-carrier access [12]. Few asked the question that matters most for real-world users: **What cellular performance does a typical user experience in the 5G era while on the move?**

In 2022, we set out to answer this question the long way, by driving more than 5,700 kilometers from Los Angeles to Boston (Figure 1a). Along the way, we measured every major US network and every available cellular technology – LTE, low-band 5G, mid-band 5G, and mmWave. We tracked performance across the protocol stack and evaluated several of the apps 5G was supposed to transform. It was the first study of its kind at this scale across the continental US.

But the 5G landscape keeps evolving. In just two years, carriers accelerated their shift from Non-Standalone (NSA) to Standalone (SA) 5G, added higher-order carrier aggregation, and activated new spectrum bands with different duplexing modes. All these changes were meant to bring 5G closer to its original vision: faster, more stable, and more reliable, even in motion. This raised a new question: **If we retraced the same journey two years later, what would we find?**

So in 2024, we did it again: same route, same methodology, same three major carriers. What follows is the first side-by-side look at cellular experience on the move in 2022 and in 2024: how the coverage landscape has changed, how cellular performance has evolved, and whether the 5G "killer apps" are any closer to becoming everyday realities.

HOW WE MEASURED CELLULAR NETWORKS ON THE MOVE

Large-scale measurements pose unique challenges and require a systematic approach to address them. Under mobility, several factors can limit performance, including operator infrastructure and policies, vehicle speed, app design, and server location. Furthermore, performing analysis from the end-user perspective adds additional complexities, as commercial devices do not expose lower-layer KPIs needed for deeper analysis, and most 5G "killer apps" are not yet commercially available. To address these challenges, we built our own testbed (Figure 1b) – a measurement platform comprising multiple smartphones, laptops, and power supplies; cloud and edge servers across the country; professional tools (XCAL Solo) for collecting lower layer KPIs and control-plane signaling; a mix of custom-built and readily available 5G "killer apps"; and a custom-built software that includes a suite of tests (throughput, RTT, application QoE) and procedures for synchronization and post-processing of logs from different software and different layers of the protocol stack.

NETWORK COVERAGE: A VERY DIFFERENT 5G MAP IN 2024

During the first driving trip in 2022, 4 years after the initial 5G rollout, we found, surprisingly, that the technology was still at an early stage. 5G coverage (Figure 2) was disappointingly low and highly fragmented across the country, with two operators (Verizon and AT&T) offering less than 25% of 5G coverage along the driving route. Moreover, the real promise of 5G – the higher capacity and faster speeds of mid-band and mmWave – was unavailable in large segments of the route. The coverage of "high-speed 5G" (mid-band or mmWave) ranged from 38% (T-Mobile) to as low as 3% (AT&T).

By 2024, the coverage landscape changed drastically for all three operators. T-Mobile, which already had a strong lead in 5G deployment in 2022, now delivered 5G along nearly the entire route. We measured 5G availability during 94% of our drive, compared to 68% in 2022. Verizon and AT&T also boosted their 5G presence, from 22% to 49% and from 18% to 45%, respectively. The high-speed 5G coverage also increased for all three operators from 2-37% in 2022 to 18-81% in 2024, with mid-band becoming the norm for two out of three operators (Verizon and T-Mobile). These numbers reflect substantial mid-band rollouts on long stretches of interstate highways and explain much of the performance improvements we describe later. Nevertheless, despite these improvements, five years after the initial 5G rollout, Verizon and AT&T still relied on LTE along considerable portions (51% & 55%) of the driving route. Additionally, mmWave remained essentially non-existent on our route in both years. Because of its range limitations and sensitivity to blockage, it still plays no meaningful role in high-mobility scenarios.

We also saw shifts in core network architecture. In 2022, practically all 5G ran in NSA mode, meaning LTE was still responsible for the control plane. By 2024, T-Mobile had rolled out widespread 5G SA (60.5% vs. only 2.5% in 2022), allowing the network to operate independently of LTE. On the other hand, Verizon and AT&T s till relied exclusively on NSA 5G.

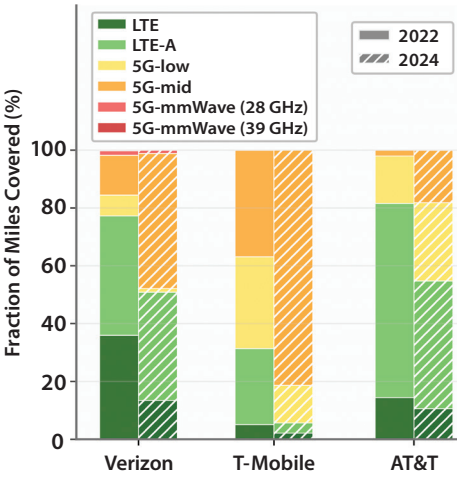


FIGURE 2. Cellular network coverage, defined as the fraction of miles driven under a given technology, in 2022 and 2024.

NETWORK PERFORMANCE: THROUGHPUT BOOSTS BUT LATENCY HOLDS BACK

Coverage improvement is only half the story. The real question is how users experience the network while on the move. To understand whether the improved 5G coverage in 2024 translates into better user experience, we next perform a head-to-head comparison of the network performance across the two years.

Figure 3 shows a clear multi-fold improvement in the throughput in 2024 in both traffic directions for all three operators. The median downlink throughput (Figure 3, left) increased from 16/6/34 Mbps in 2022 to 148/248/77 Mbps in 2024 for Verizon, T-Mobile, and AT&T, respectively. Similarly, the median uplink throughput (Figure 3, middle) increased from 9/6/8 Mbps to 23/35/18 Mbps. These large throughput improvements align with the increasing presence of mid-band 5G, which provides the combination of range and capacity needed to sustain high bitrates on highways. Whether we were passing through major cities or rural areas, the impact of mid-band 5G was obvious, particularly for Verizon and T-Mobile, whose 5G coverage is dominated by 5G-mid band deployments (Figure 2): *2024 finally delivered the fast speeds that early 5G marketing had promised.*

On the other hand, the latency (Figure 3, right) tells a different story. If 5G were living up fully to its low-latency expectations, here we would show sharply lower RTTs in 2024. Instead, improvements were modest, and sometimes absent. For Verizon and T-Mobile, latency remained roughly flat or slightly worse in some regions, even where throughput improved. AT&T was the exception, showing a meaningful reduction in RTT (about 10 ms), but even then, the gains were not transformative. This reflects the reality that latency depends on far more factors than radio technology. While lower layer innovations such as MIMO, increased levels of carrier aggregation, and the use of wider channels, or improvements in the modem and signal processing capabilities can directly impact throughput, the end-to-end latency is affected by multiple factors including scheduling policies at the base station, prioritization policies (of ICMP vs. TCP traffic), as well as congestion both in the operator's network and the end-to-end path. In short, *2024's 5G is much faster, but not quicker.*

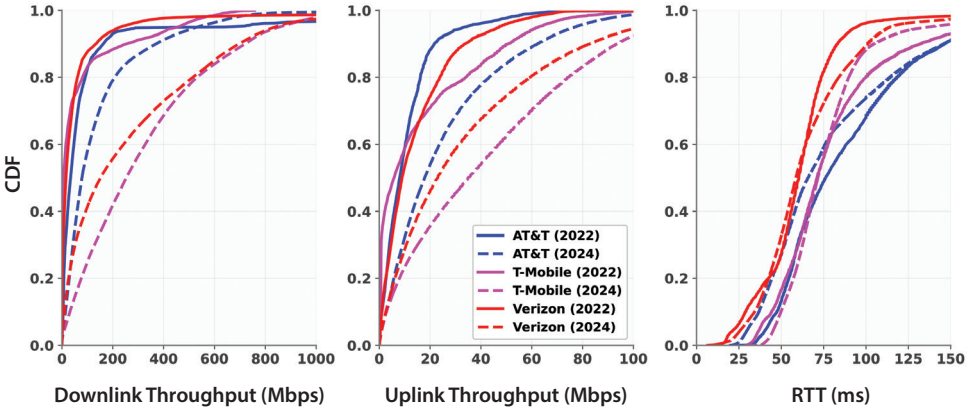
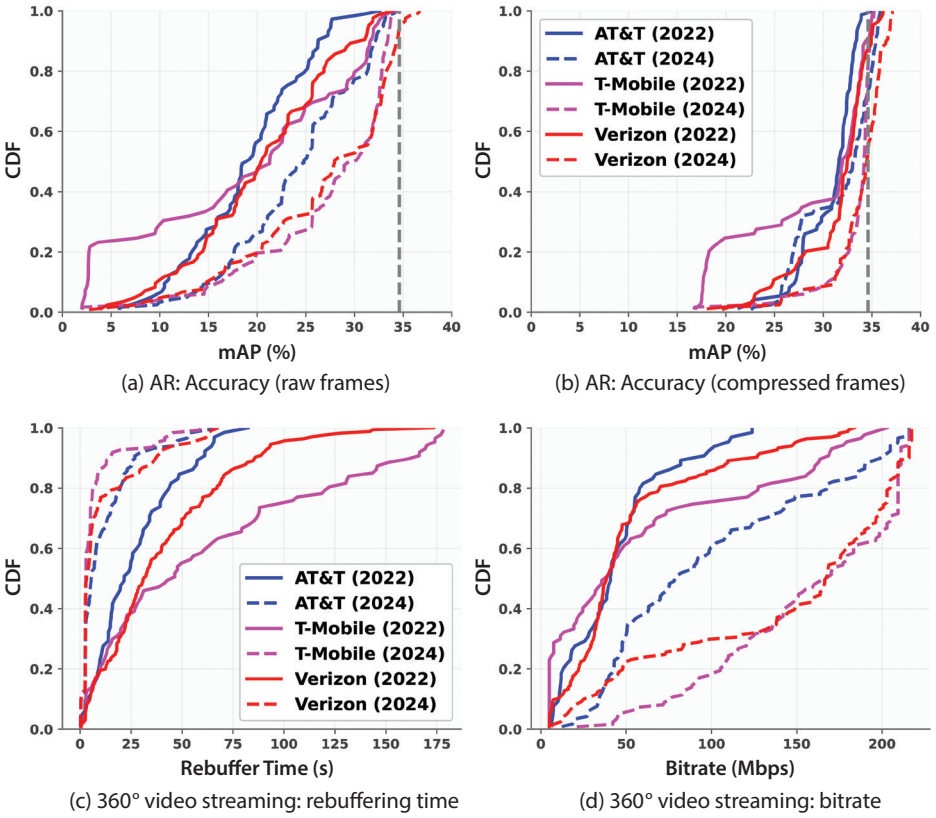


FIGURE 3. Throughput and RTT comparison of all three operators in 2022 and 2024.



(a) AR: Accuracy (raw frames)
(b) AR: Accuracy (compressed frames)
(c) 360° video streaming: rebuffering time
(d) 360° video streaming: bitrate

FIGURE 4. Comparison of the performance of 5G "killer apps" in 2022 vs. 2024.

PUTTING 5G TO THE TEST: TWO REAL APPLICATION CASE STUDIES

Throughput and latency measurements are useful, but the real test of network maturity comes from actual applications. To assess that, we examined two representative 5G "killer apps": cloud-assisted AR (uplink-centric) and 360° video streaming (downlink-centric).

Cloud-Assisted AR. We built a canonical cloud-assisted AR app that offloads camera frames, captured at a rate of 30 FPS, to a cloud/edge server with an NVIDIA A100 GPU for DNN-based object detection. The app has two versions: one offloading uncompressed frames of 450 KB, and another one offloading compressed frames of 50 KB. Both versions use local tracking [14] to mask long end-to-end latencies and improve the object detection accuracy based on stale results from the server. We evaluated its performance by measuring the object detection accuracy (mAP metric [2]). Similar to previous work [3], we consider the performance to be satisfactory if the mAP with offloading is within 90% of the offline accuracy (shown as a dashed line in Figures 4a, 4b).

In 2022, cellular networks could not support cloud-assisted AR under driving. Even with compression (Figure 4b), the median mAP was 31.6-32.5% (82-85% of the offline mAP). The performance was much worse without compression (Figure 4a), only 19.2-21.3% (50-55% of the offline accuracy). By 2024, things improved significantly. With mid-band coverage far more abundant, the median mAP with raw frames increased to 24.8-30.1% (64-78% of the offline accuracy). The improvement was even more pronounced with compressed frames; the median mAP increased to 33.3-34.5% (87-90% of the offline accuracy).

Our results also show that application-level optimizations developed in the pre-5G era remain relevant even in 2024, despite the drastic improvements in cellular throughput. Compression, combined with local tracking, which helps mask the long end-to-end latency, helped the app achieve satisfactory accuracy 25-50% of the time in 2024 despite offloading at most 12.5 FPS instead of 30 FPS. However, the accuracy still remained unsatisfactory for a large fraction of time for all three operators, especially during periods of low-band 5G or LTE coverage, and sensitive to latency spikes (recall from Figure 3 that latency has shown little to no improvement in 2024). Overall, *the experience in 2024 was viable, but far from flawless.*

360° video streaming. Our 360° video streaming app is based on Puffer [1], an open-source video streaming server, and our custom-built client. The app used 2-second segments encoded at 4 different bitrates (100, 50, 10, and 5 Mbps) in 2022 and 5 bitrates (250, 100, 50, 10, and 5 Mbps) in 2024.

The results in Figures 4c and 4d tell a more optimistic story compared to the AR use case, as video streaming is highly sensitive to downlink throughput but much less dependent on latency, unlike AR. In 2022, the average session bitrate was only 40 Mbps in the median case and video stalls were very common and excessively long. Median rebuffering times ranged from 22-45% of the session time and one quarter of the sessions over T-Mobile experienced a rebuffering time of 100%. In 2024, the performance improved drastically, thanks to the widespread mid-band 5G deployment, which led to a drastic improvement in downlink bandwidth. The average bitrate increased to 79-168 Mbps and stalls became much less frequent; 48-71% of the session experienced rebuffering for less than 5% of the time.

CONCLUSION AND THE ROAD AHEAD

The results from the two driving trips clearly show that the US mobile landscape has undergone a real transformation in just two years, from 2022 to 2024. Mid-band 5G finally reached highways, and with it came the kind of sustained throughput that 5G "killer apps" like cloud-assisted AR or 360° video streaming depend on. 2024 mobile networks deliver a fundamentally different experience for users of such apps than their 2022 counterparts.

But the story is not uniformly bright. Latency remains the biggest obstacle preventing truly seamless experience for latency-sensitive apps such as AR or cloud gaming. Even with SA deployments emerging, several pieces of the puzzle that are needed to bring down the end-to-end

latency are still maturing. mmWave continues to play no meaningful role for users on highways, and the situation will likely not change in the near future. And despite substantial improvements in available bandwidth, app developers still need to rely on application-level optimizations developed in the pre-5G era to tolerate jitter, handle coverage transitions gracefully, and fall back intelligently during performance dips.

The good news is that the foundation is finally here. In 2024, the networks have the raw bandwidth to support the new generation of mobile apps. With smarter application design and continued investment in low-latency infrastructure, the next few years may finally bring the seamless, high-performance experience that 5G has long promised. The road ahead certainly looks far more promising than it did just two years ago. ■

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